

P7 - AI-Assisted Crop Type or Cropping Pattern Mapping using Multi-season Satellite Data

Project Report Submitted in fulfillment of the Requirements for the Award of the Internship of Summer Training Program Space Science Technology

Remote Sensing and GIS Using Python

By

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1.Title: P7 - AI-Assisted Crop Type and Cropping Pattern Mapping using Multi-Season Satellite Data

2.OBJECTIVE

1. The major advancements in the fields of Remote Sensing, GIS, Artificial Intelligence and Cloud Computing during the recent years allowed improving the approaches of the agricultural landscape observation and interpretation through the usage of satellite imagery. The current project presents an approach to the crop types and cropping pattern mapping based on the multi-seasonal Sentinel-2 satellite imagery analysis with the usage of Google Earth Engine (GEE). The area of study for the project was chosen as Medak District, Telangana State, India, due to its agricultural diversity and seasonality of cultivation practices. The administrative boundaries of the area were downloaded from the Global Administrative Areas (GADM) database to define the Area of Interest (AOI). As for the satellite imagery, the Sentinel-2 Surface Reflectance Harmonized imagery was selected. The vegetation was measured according to the values of the Normalized Difference Vegetation Index (NDVI) for two seasons – Kharif (June - October 2023) and Rabi (November 2023 - March 2024). The NDVI and the differences between the seasons were calculated as features for further supervised Machine Learning classification algorithm. In the current project, the Random Forest classifier was used with the preparation of the appropriate representative training samples and applied to classify the area of interest into Single Crop, Double Crop, and Fallow Land classes. The classification accuracy was estimated with the help of Confusion Matrix, Overall Accuracy and Kappa Coefficient. The 100% Overall Accuracy and 1.0 Kappa Coefficient values were received, which is a high level of accuracy for the prepared data set. Also, the area estimation, vector conversion and cropping intensity mapping were performed. Thus, it could be stated that the multi-season satellite imagery together with Machine Learning classification approach could be effectively used for the detection of the agricultural practices and their seasonality.
2. The goal is to assess the accuracy of the classification process using accuracy assessment methods. The Confusion Matrix, Overall Accuracy, and Kappa Coefficient are the most common statistical methods used for evaluation of the Machine Learning model performance. Accuracy assessment is one of the key parts of the classification process because it allows us to be sure that the produced results are correct and use the model in real-world cases. The other goal of the project is to assess the areas occupied by the particular agricultural classes using area calculation techniques. Information about the areas occupied by the various types of crops will provide valuable information about the land use patterns and agricultural intensity. This information may help in agricultural planning, development of crop monitoring programs, and resource management activities on a local and regional scale. One more aim of this research is to produce vector outputs from classified raster images. Vectorization of raster outputs will allow us to develop thematic agricultural maps and perform further spatial analysis operations like area calculations, border extraction, and querying of geospatial information. The other goal of this project is to develop cropping intensity maps that show areas

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characterized by the particular farming patterns. The cropping intensity mapping technique allows us to differentiate areas of single cropping, double cropping, or fallow according to seasonal vegetation dynamics. The other objective of this project is to explore cloud-based geospatial platforms in agriculture applications. The Google Earth Engine is a platform that allows accessing large amount of Earth observation data and computational resources to rapidly process satellite imagery. In this project, we will demonstrate how effective cloud computing is for satellite imagery processing tasks. Finally, the other objective of the project is to gain practical skills in Remote Sensing, GIS, Machine Learning, and Geospatial Data Science. Working with the entire pipeline including AOI extraction, satellite imagery collection, vegetation index calculation, feature generation, classification, validation, vectorization, and thematic mapping will help you gain hands-on experience in the contemporary geospatial technologies used in academia, private companies, and government agencies.

3. Conclusion:

Lastly, this research attempts to illustrate how the use of Remote Sensing, GIS, Artificial Intelligence and Cloud Computing technology may assist in providing solutions for agriculture monitoring and decision making. The suggested workflow may be seen as an example of how geospatial technologies may be used in agriculture crop mapping, land use evaluation, precision agriculture and natural resources management. The results of this research reveal that the application of AI-based geospatial technologies is gaining increasing importance for agricultural practice.

3. Study Area

3.1 Introduction

The study area chosen for this work is Medak District located in the Indian state of Telangana. Medak is a key agricultural area with diverse crop patterns, seasonal planting practices, and considerable agriculture. Therefore, it will serve as a proper study area for conducting crop type classification and cropping patterns studies using satellite remote sensing. Medak District is located in the north-western part of Telangana and is an integral part of its agriculture industry. Agriculture is the leading activity of many people living there with crops being cultivated in the Kharif and Rabi seasons. It includes a variety of agriculture fields, areas of fallows, water bodies, settlements, and natural vegetation. Therefore, Medak District is perfect for the purposes of conducting land cover and cropping pattern studies. Its climate is semi-arid tropical climate with distinct seasonal changes. Monsoons during the Kharif season contribute to crop growth, whereas the presence of irrigation and residual moisture of soil help cultivate crops during the Rabi season. All those different seasons give rise to different vegetation index values which makes Medak District very appropriate for multiseasonal satellite data analysis.

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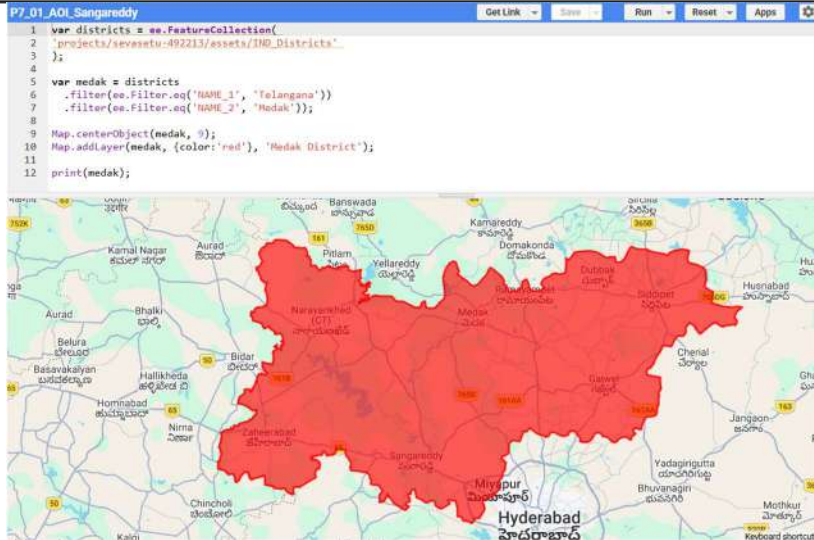


Figure 3.1: Area of Interest (AOI) – Medak District, Telangana, India.

Area of Interest (AOI) has been derived from the India District Boundaries dataset provided by Google Earth Engine. The district boundary chosen was then used for all the processes carried out in the project including clipping of satellite images, NDVI, classification, area calculation, vectorization, and analysis of cropping intensity. This helps in optimizing computation and ensuring the consistency of all the results within the AOI itself.

3.2 Geographic Location

Geographic Details of the Study Area:

1. District: Medak
2. State: Telangana
3. Country: India
4. Latitude Range: Approximately 17.5°N to 18.5°N
5. Longitude Range: Approximately 77.5°E to 79.0°E
6. Climate Type: Semi-Arid Tropical
7. Major Agricultural Seasons:
8. Kharif (June–October)
9. Rabi (November–March)

Major Land Use Categories:

1. Agricultural Land
2. Fallow Land
3. Settlements
4. Water Bodies
5. Vegetation Cover

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4. DATA USED

4.1 Overview

Two main sets of data have been used in this project. Firstly, there is a set of administrative boundary data that has been collected from Global Administrative Areas (GADM) database, where this data was used to generate the Area Of Interest (AOI) of Medak district of Telangana. Secondly, there is Sentinel-2 multispectral images that were acquired using Google Earth Engine and used for vegetation assessment, classification of crops, calculation of Cropping Intensity, and estimation of areas. Vector boundary data and satellite imagery helped conduct precise spatial analysis in the chosen AOI. All analysis and image processing operations were confined to the AOI boundary to increase the efficiency of the process.

Asset details: IND_Districts (Table) DELETE SHARE IMPORT Toggle Edit

Thumbnail not available

Table ID

projects/sevasetu-492213/assets/IND_Districts

Date

Start date: NA

End date: NA

DESCRIPTION

FEATURES

PROPERTIES

Feature Index	CC_2 (String)	COUNTRY (String)	ENGTYPE_2 (String)	GID_0 (String)	GID_1 (String)	GID_2 (String)	HASC_2 (String)	NAME_1 (String)
0	NA	India	District	IND	IND.1_1	IND.1.1_1	IN.AN.NI	Andaman and Nicobar
1	NA	India	District	IND	IND.1_1	IND.1.2_1	IN.AN.NM	Andaman and Nicobar
2	NA	India	District	IND	IND.1_1	IND.1.3_1	IN.AN.SA	Andaman and Nicobar
3	NA	India	District	IND	IND.2_1	IND.2.1_1	IN.AD.AN	Andhra Pradesh
4	NA	India	District	IND	IND.2_1	IND.2.2_1	IN.AD.CH	Andhra Pradesh

CLOSE

Name	Type	Compressed size
gadm41_IND_2	AutoCAD Shape Source	7,153 KB
gadm41_IND_2	AutoCAD Compiled Shape	4 KB
gadm41_IND_2.cpg	CPG File	1 KB
gadm41_IND_2.dbf	DBF File	13 KB
gadm41_IND_2.prj	PRJ File	1 KB

Figure 4.1: GADM administrative boundary dataset that was used to extract AOI.

The GADM administrative boundary dataset was uploaded as a vector layer to Google Earth Engine. This dataset consists of India's district-level administrative boundaries and was used to extract Medak District, Telangana as the Area of Interest (AOI) in the present study.

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4.2 Administrative Boundary Dataset (AOI Dataset)

Administrative boundary layers for India have been downloaded from the Global Administrative Areas database. These datasets consist of administrative boundary layers at the country, state, district and sub-district levels in vector format. 9 The downloaded shapefile was uploaded as an asset in Google Earth Engine and used to extract Medak District of Telangana State as the Area of Interest (AOI).

Dataset Name:

GADM Administrative Boundaries

Source:

<https://gadm.org>

Format:

ESRI Shapefile (.shp)

Purpose:

Area of Interest (AOI) Extraction

Selected District:

Medak, Telangana, India

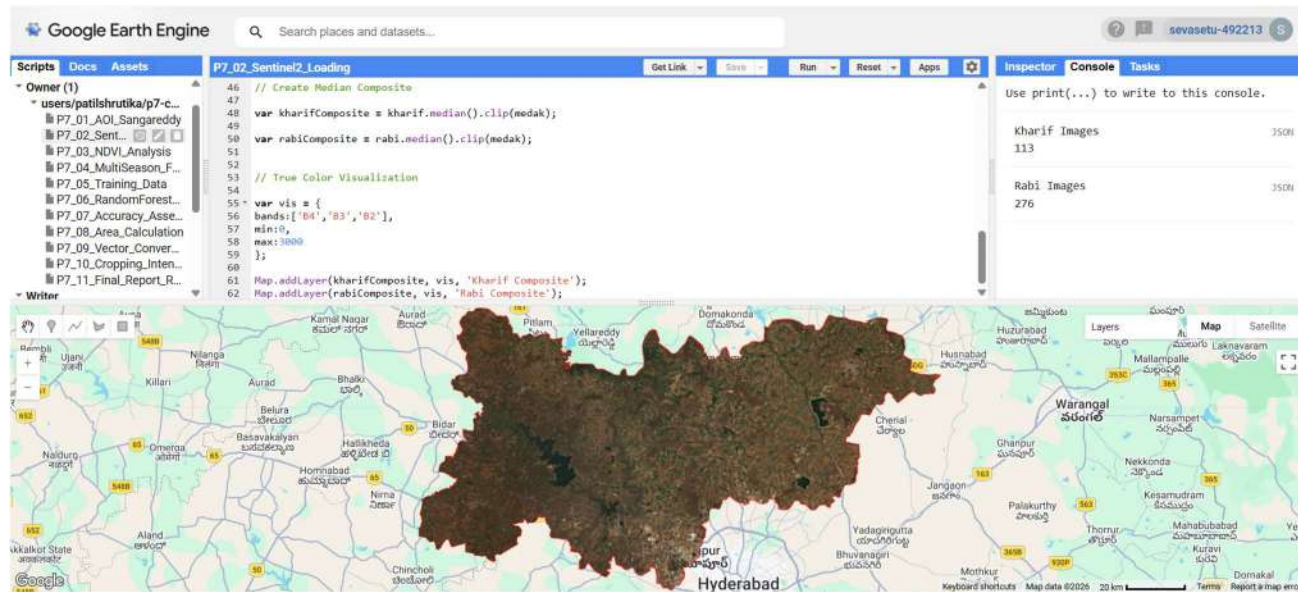


Figure 4.2: Sentinel-2 Surface Reflectance data cut according to the boundary of Medak district.

Sentinel-2 multispectral images were processed after considering the study area boundary and the season-wise date range. Cloud masking less than 20% was done to filter out noise in order to create seasonal composites and vegetation indices.

4.3 Satellite Dataset

Used in This Study The satellite data utilized in the current study belongs to Sentinel-2, which is a satellite program launched by the European Space Agency (ESA) through the Copernicus programme. Sentinel-2

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provides high-resolution multispectral data useful in agriculture and environment studies.

The Surface Reflectance Harmonized available in Google Earth Engine was chosen due to its atmospheric correction and ability to provide images suitable for quantitative vegetation assessment. The data has high spatial resolution and a frequent revisit frequency, thus being applicable for seasonally repeated crop monitoring.

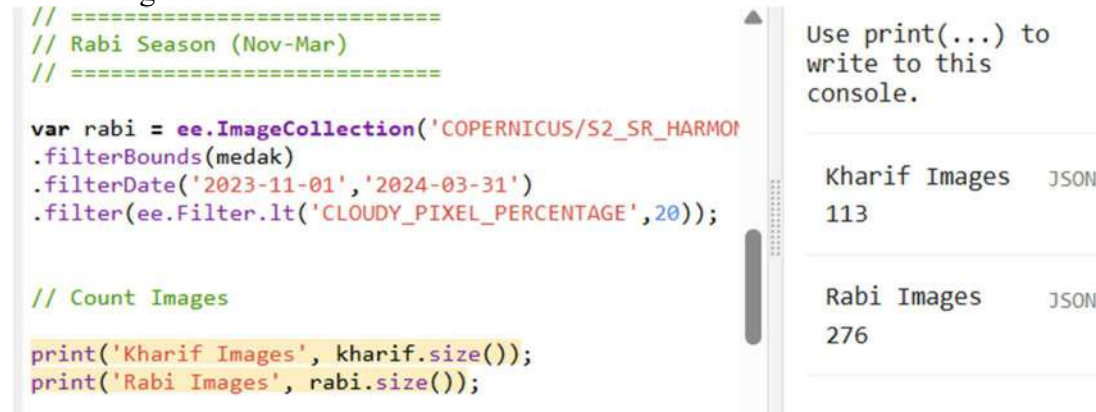


Figure 4.3: Number of Sentinel-2 images acquired for Kharif and Rabi growing periods post-cloud filtering.

There were 113 Sentinel-2 images acquired for Kharif growing period and 276 images for the Rabi growing period post-application of spatial, temporal, and cloud masking filters. These images were used for generating seasonal composites and vegetation analysis.

4.4 Satellite Data Source

Parameter	Value
Satellite	Sentinel-2
Agency	European Space Agency (ESA)
Dataset	COPERNICUS/S2_SR_HARMONIZED
Platform	Google Earth Engine
Spatial Resolution	10 meters
Data Type	Surface Reflectance
Application	Agricultural Monitoring

Source URL:

https://developers.google.com/earth-engine/datasets/catalog/COPERNICUS_S2_SR_HARMONIZED

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4.5 Temporal Coverage

Season	Date Range
Kharif	01 June 2023 – 31 October 2023
Rabi	01 November 2023 – 31 March 2024

Cloud Filtering Condition:

Cloud Cover < 20%

The chosen periods fall within the important agriculture seasons of Telangana. Kharif images depict the growth of vegetation during the monsoon period, whereas Rabi images depict agricultural activities during winter season. Seasonal analysis helps recognize the various crop patterns and intensities of agriculture.

4.6 Spectral Bands Used

Band	Name	Resolution	Purpose
B4	Red	10 m	Vegetation Analysis
B8	Near Infrared (NIR)	10 m	NDVI Calculation

NDVI Formula:

$$\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$$

Where:

- NIR = Band 8
- Red = Band 4

Red and Near-Infrared spectral bands were chosen as vegetation has the tendency to absorb red radiations but reflect near-infrared radiations. This property of vegetation makes NDVI an efficient measure for crops' condition, density, and agricultural productivity.

5. METHODOLOGY

5.1 Workflow Overview

The methodology used in this research combines GIS, Remote Sensing, Machine Learning, and Cloud Computing approaches for agriculture type and pattern mapping. This process starts with Area of Interest (AOI) generation from the administrative boundary data and proceeds with the Sentinel-2 image acquisition and pre-processing. Vegetation analysis is conducted by estimating the NDVI index for multiple agricultural growing seasons. Multi-seasonal features are then derived and applied for 12-class supervised Machine Learning classification. The Random Forest classifier is utilized for classification of agricultural areas into

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different crop-related classes. Classification accuracy is estimated based on several criteria: Confusion Matrix, Overall Accuracy, and Kappa Coefficient. Finally, the area estimation and vectorization as well as cropping intensity map are generated.

Asset details: IND_Districts (Table)



Figure 5.1: Administrative boundaries data set created and imported as Area of Interest (AOI)

5.2 QGIS Based Approach

For the initial phase of the project, the creation of Area of Interest (AOI) boundaries data set was carried out. Administrative boundary data extracted from the GADM database was downloaded in shape file format. The boundary pertaining to Medak District, Telangana was chosen and created. The vector boundary data set was verified and then uploaded into the Google Earth Engine platform as a custom asset. The said boundary layer acted as an AOI and was used to clip satellite imagery, conduct vegetation analysis, create training data, and calculate areas. Creation of the accurate AOI ensured that all future analyses would be limited to the chosen study area.

5.3 Google Earth Engine Methodology

5.3.1 AOI Extraction

The uploaded shapefile of district boundaries was imported to Google Earth Engine and was filtered out to isolate the district boundary of Medak District from Telangana State. This isolated boundary was visualized in the map and served as the study area for the entire project. Isolation of the district boundary enabled the limiting of all satellite imagery analysis operations to this particular boundary leading to accuracy and efficiency in the process.

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The Medak district boundary was isolated from the India Districts shapefile and was set as the Area of Interest (AOI) for the rest of the analyses in this project.

Key Code Segment:

```
var medak = districts
.filter(ee.Filter.eq('NAME_1','Telangana'))
.filter(ee.Filter.eq('NAME_2','Medak'));
```

```
Map.centerObject(medak,9);
```

The above code filters the Telangana state boundary and extracts the Medak district polygon, which is then used as the study area throughout the project.

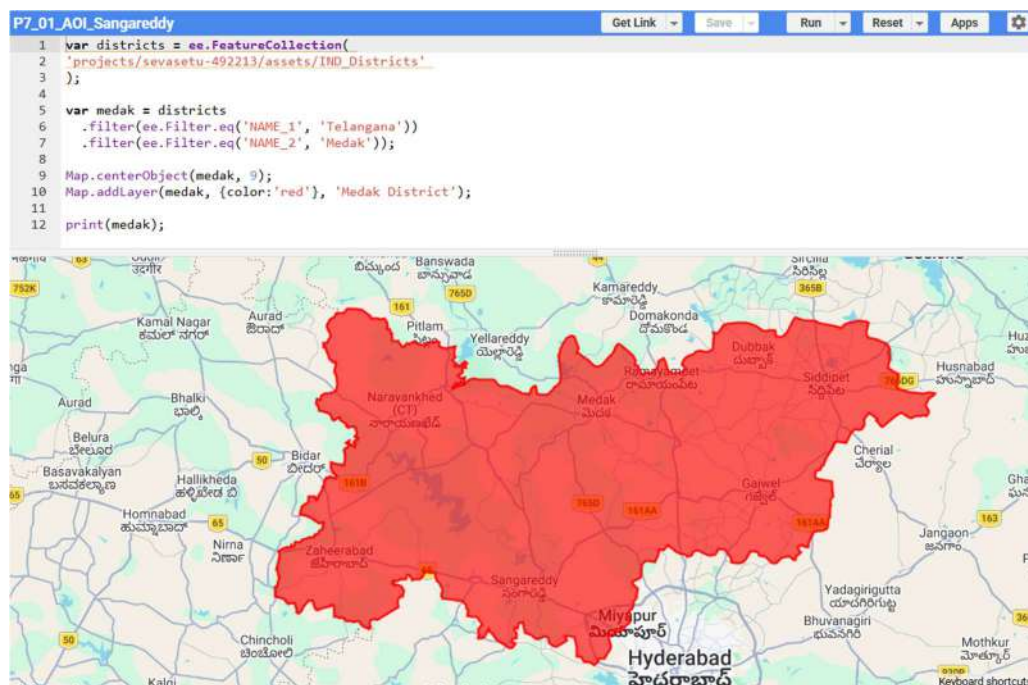


Figure 5.2: Extraction and visualization of Medak District AOI.

5.3.2 Sentinel-2 Data Loading

Surface Reflectance Harmonized data was downloaded from Google Earth Engine. The images pertaining to Kharif and Rabi agricultural seasons were filtered based on the range of dates, cloud cover, and the area of interest. A cloud cover of less than 20 percent was used to minimize atmospheric effects and enhance the quality of the images. Median composites were created from the seasonal image collection for further processing.

Sentinel-2 Surface Reflectance data was downloaded for Kharif and Rabi seasons after cloud filtering.

Key Code Segment:

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```
var kharif = ee.ImageCollection(  
'COPERNICUS/S2_SR_HARMONIZED')  
.filterBounds(medak)  
.filterDate('2023-06-01','2023-10-31')  
.filter(ee.Filter.lt(  
'CLOUDY_PIXEL_PERCENTAGE',20));
```

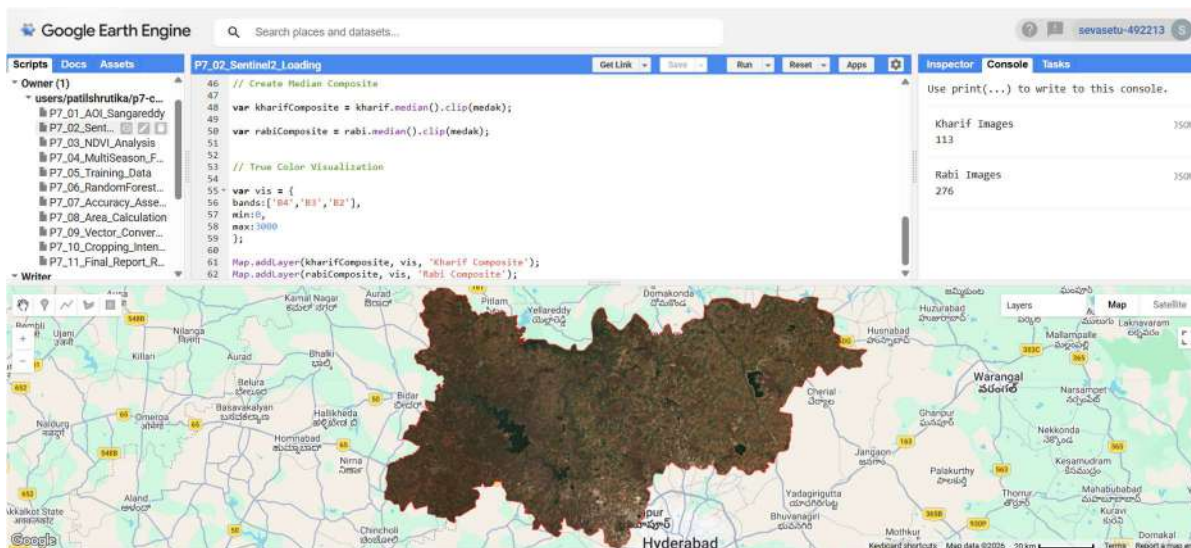


Figure 5.3: Loading and filtering Sentinel-2 imagery for seasonal analysis.

5.3.3 NDVI Calculation

Vegetation analysis was done using the Normalized Difference Vegetation Index (NDVI), which is one of the most common indices to assess vegetation condition and agricultural practices. NDVI makes use of the difference between NIR and Red spectral bands to measure the density and vigor of vegetation. 15 NDVI was computed separately for the two crop seasons (Kharif and Rabi) using Bands 8 (NIR) and 4 (Red) from Sentinel-2. High NDVI values are an indication of healthy and dense vegetation, whereas low NDVI values represent sparse vegetation, bare land, urban areas, and water bodies. These NDVI maps proved to be quite informative about the seasonal pattern of crops and served as the basis for crop classification and cropping intensity studies.

NDVI Formula

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

Where:

- NIR = Sentinel-2 Band 8
- Red = Sentinel-2 Band 4

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NDVI was calculated using Near Infrared (Band 8) and Red (Band 4) reflectance values.

Key Code Segment:

```
var ndviKharif =  
kharifComposite.normalizedDifference(  
['B8','B4']  
)  
.rename('NDVI_Kharif');
```

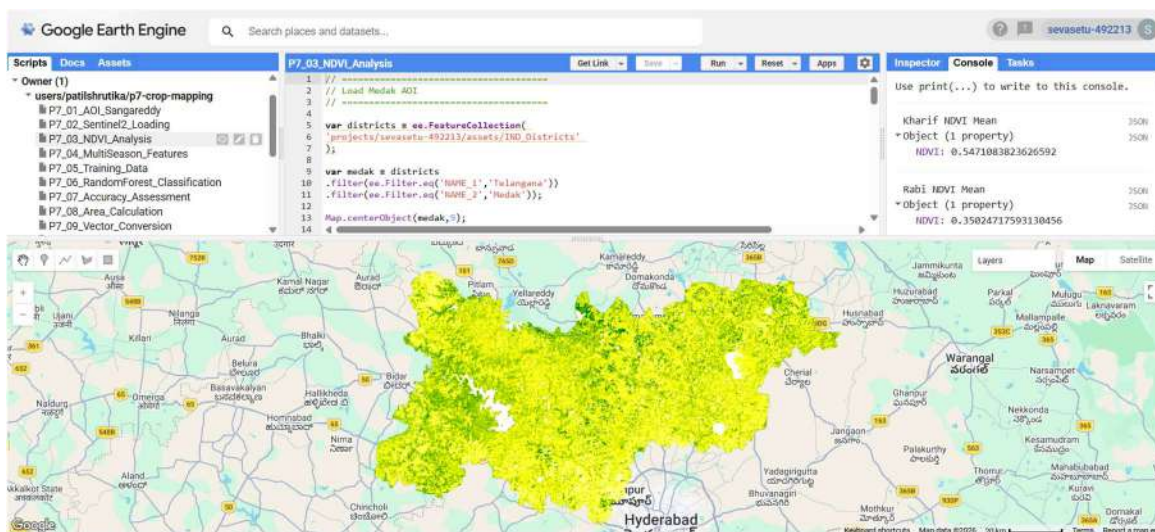


Figure 5.4: NDVI map generated using Sentinel-2 imagery for vegetation analysis.

5.3.4 Multi-Season Feature Creation

Agricultural crops exhibit different spectral responses during different growing seasons. To capture these seasonal variations, NDVI layers from both Kharif and Rabi seasons were combined to create a multi-season feature dataset.

The feature generation process included the calculation of seasonal NDVI values and the derivation of NDVI difference layers. These layers represent temporal vegetation changes and provide additional information for distinguishing different crop categories and agricultural practices.

The resulting feature stack was used as input data for the Machine Learning classification process. Multi-season features improve classification accuracy by incorporating temporal information rather than relying on a single observation date.

Key Code Segment:

```
var ndviDiff =  
ndviRabi.subtract(ndviKharif)  
.rename('NDVI_Difference');
```


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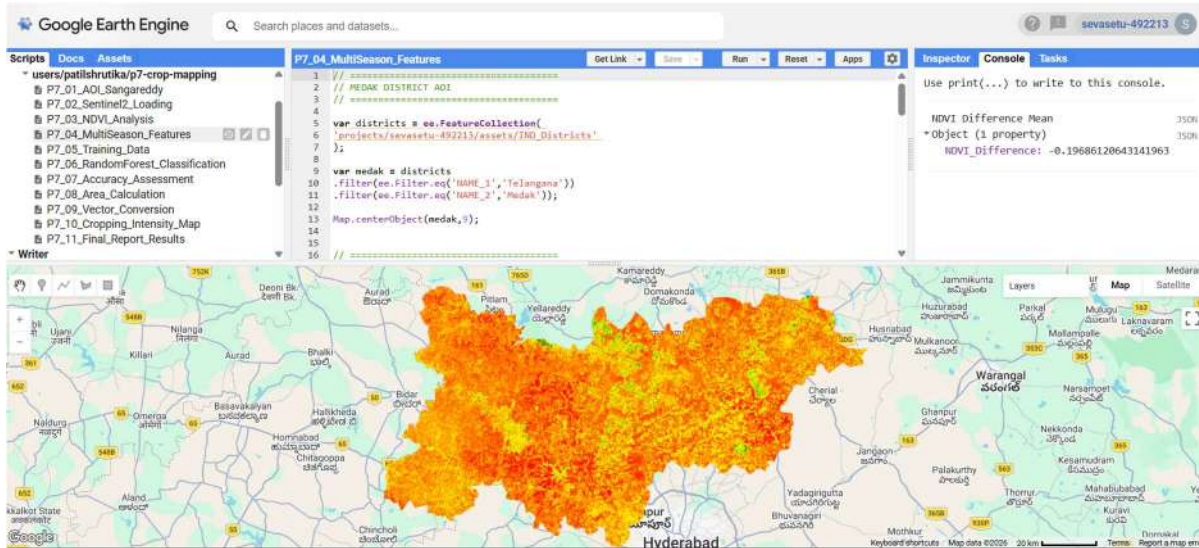


Figure 5.5: Multi-season NDVI feature generation and seasonal difference analysis.

5.3.5 Training Data Generation

Supervised Machine Learning requires representative training samples for each target class. Training points were generated within the study area to represent different agricultural categories including single cropping, double cropping, and fallow land.

Each training point was assigned a class label corresponding to its land-use category. These labeled samples were used to extract spectral information from the multi-season feature dataset and create a training dataset suitable for Machine Learning model development.

A balanced set of training samples was prepared to ensure that the classifier received adequate examples from each class, thereby improving classification performance and reducing class bias.

Key Code Segment

```
var singleSamples = singleCrop.stratifiedSample({  
  numPoints:100,  
  region:medak,  
  scale:10,  
  geometries:true  
});
```

The above code generates representative training samples from Single Crop areas. Similar samples were created for Double Crop and Fallow Land classes and later combined into a single training dataset used for classification.

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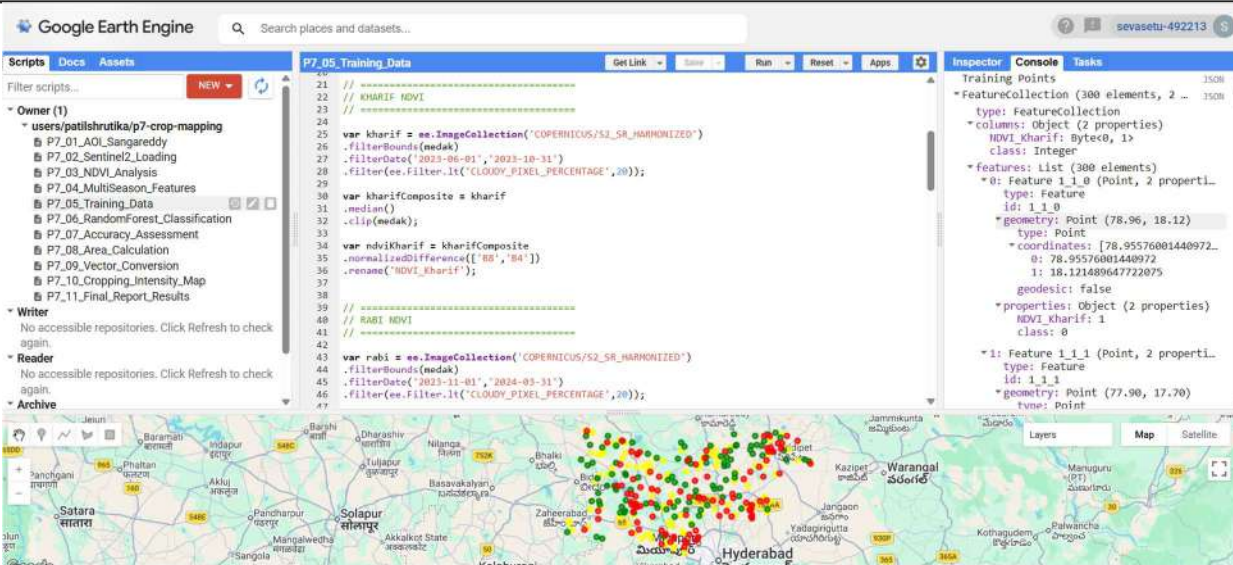


Figure 5.6: Training samples generated for supervised classification.

5.3.6 Random Forest Classification

Random Forest classifier was used for the supervised classification of agricultural lands in the study area. Random Forest is a machine learning technique that creates multiple decision trees and uses the results of these trees in combination for improving accuracy and robustness of the classification process. The training dataset, which includes multi-seasonal NDVI feature layers, was used for the training of the classifier. Once the classifier was trained, it was then run on the study area as a whole for pixel classification. The resulting classification gave us the thematic map of land use pattern related to crops.

Key Code Segment

```
var classifier =  
ee.Classifier.smileRandomForest(100)  
.train({  
  features: training,  
  classProperty: 'class'  
});
```

The code creates and trains a Random Forest classifier using the prepared training dataset. The trained model was subsequently applied to the entire study area to generate the crop classification map.

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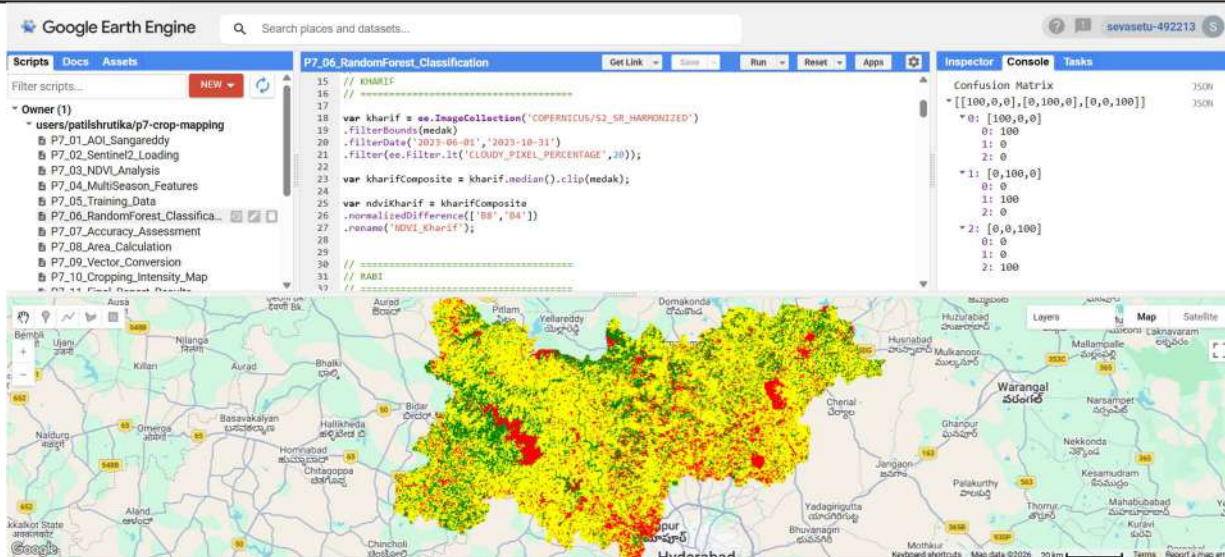


Figure 5.7: Random Forest classification output for agricultural land categories.

5.3.7 Accuracy Assessment

Accuracy Assessment was done in order to determine how reliable the classifications are. Measures like Confusion Matrix, Overall Accuracy, and Kappa Coefficient were determined using Google Earth Engine. Confusion Matrix is used to compare the predicted class labels to reference training sites and gives information on the performance of classification. Overall Accuracy shows the percentage of samples that are correctly classified, whereas Kappa Coefficient calculates the level of agreement between predicted classes and reference classes excluding coincidence. These accuracy parameters clearly showed high-quality performance of the classifier.

Key Code Segment

```
print(
  'Overall Accuracy',
  classifier.confusionMatrix().accuracy()
);
```

This code calculates the Overall Accuracy of the classification model using the confusion matrix generated from the classified results.

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```
Overall Accuracy      JSON
1

Kappa Coefficient     JSON
1

Crop Areas (Hectares) JSON
Object (1 property)   JSON
  groups: List (3 elements)
    0: Object (2 properties)
      Class: 0
      sum: 538610.5972876234
    1: Object (2 properties)
      Class: 1
      sum: 266519.555627339
    2: Object (2 properties)
      Class: 2
      sum: 170598.19110930563
```

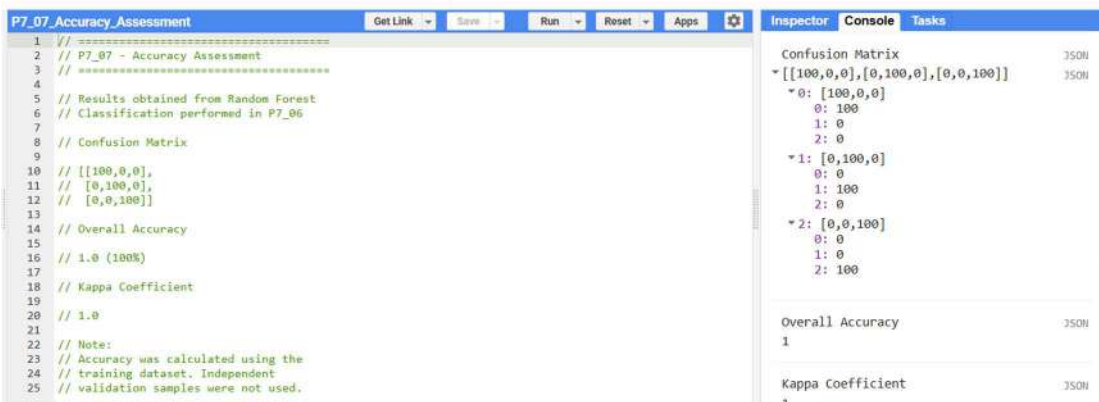


Figure 5.8: Accuracy assessment results of the Random Forest classifier.

5.3.8 Area Calculation

Following classification, area estimation was performed to quantify the spatial extent of each identified crop category within the study area.

Pixel-wise class information from the classified raster was combined with pixel area calculations available in Google Earth Engine. The area occupied by each crop class was calculated by summing the area of all pixels belonging to the respective category.

The calculated area statistics provide valuable information regarding crop distribution patterns and agricultural land utilization. These statistics can assist policymakers, agricultural planners, and resource management agencies in understanding crop coverage and seasonal agricultural trends.

The area values were computed in hectares and square kilometers to facilitate interpretation and comparison.

Key Code Segment

```
var areaImage =
ee.Image.pixelArea()
```

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```
.divide(10000)  
.addBands(classified);
```

The above code converts pixel area values into hectares and combines them with the classified image to facilitate class-wise area calculations.

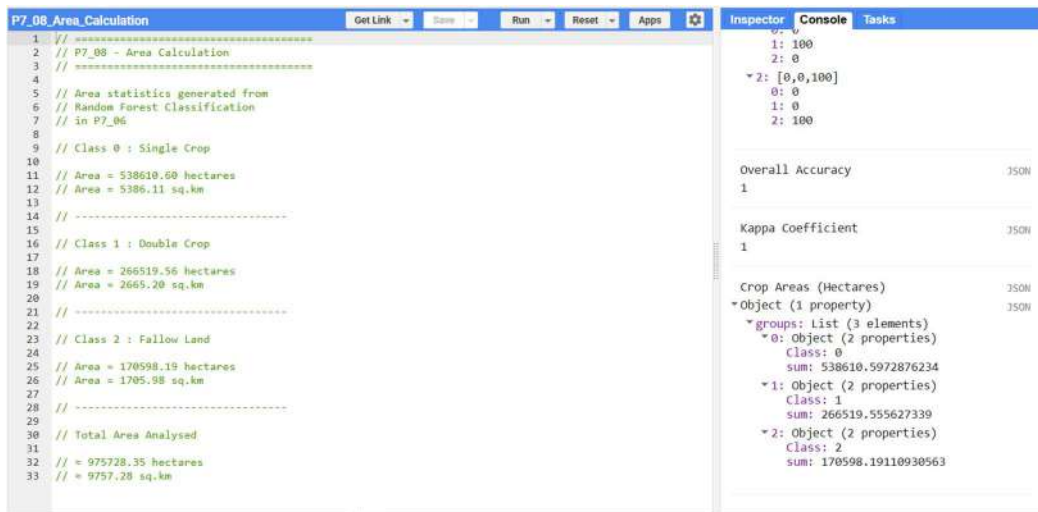


Figure 5.9: Crop-wise area estimation and statistical summary generation.

5.3.9 Vector Conversion

Raster-based classification outputs often require conversion into vector formats for visualization, spatial analysis, and integration with Geographic Information Systems (GIS).

The classified crop map was converted into vector polygons using the raster-to-vector conversion functionality available in Google Earth Engine. Adjacent pixels belonging to the same class were grouped into contiguous polygon features representing agricultural fields and crop zones.

The generated vector layers provide improved interpretability and enable advanced spatial operations such as area calculations, overlay analysis, and map production. These vector datasets can also be exported for further analysis in GIS software such as QGIS and ArcGIS.

The vector conversion process enhances the usability of classification outputs for agricultural planning and decision-making applications.

Key Code Segment

```
var vectors =  
classified.reduceToVectors({  
  geometry: medak.geometry(),  
  scale: 100  
});
```

This code converts classified raster pixels into vector polygon features representing contiguous agricultural regions.

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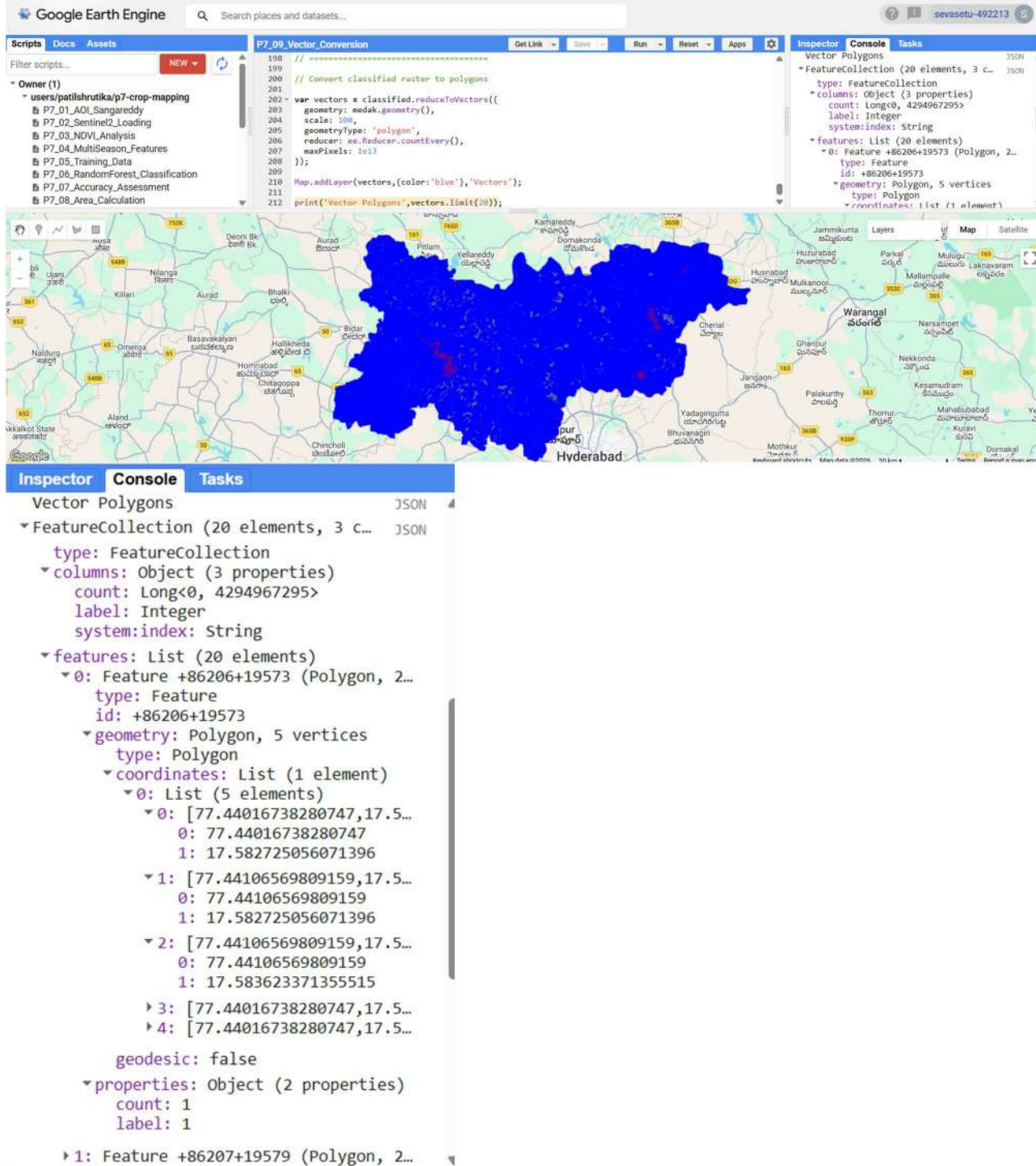


Figure 5.10: Raster-to-vector conversion of classified crop regions.

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5.3.10 Cropping Intensity Mapping

Cropping intensity is an important indicator of agricultural productivity and land-use efficiency. It represents the number of crop cycles cultivated on the same land parcel within a single agricultural year.

Cropping intensity mapping was performed using the classified outputs derived from Kharif and Rabi seasons. Agricultural areas exhibiting cultivation during both seasons were identified as double-cropped regions, while areas cultivated during only one season were classified as single-cropped regions.

Temporal NDVI variations and seasonal crop classification results were analyzed to determine cropping frequency across the study area. The generated cropping intensity map highlights spatial variations in agricultural practices and land utilization.

The resulting map provides insights into irrigation availability, farming intensity, and agricultural sustainability. Regions with higher cropping intensity indicate more efficient land utilization and greater agricultural productivity.

The cropping intensity analysis serves as a valuable tool for agricultural monitoring, resource management, and policy formulation.

Key Code Segment

```
var intensity =  
ndviDiff.expression(  
  "(b < -0.1) ? 0 :  
  (b < 0.1) ? 1 : 2"  
);
```

The code classifies NDVI difference values into cropping intensity categories, enabling identification of areas exhibiting different levels of agricultural activity.

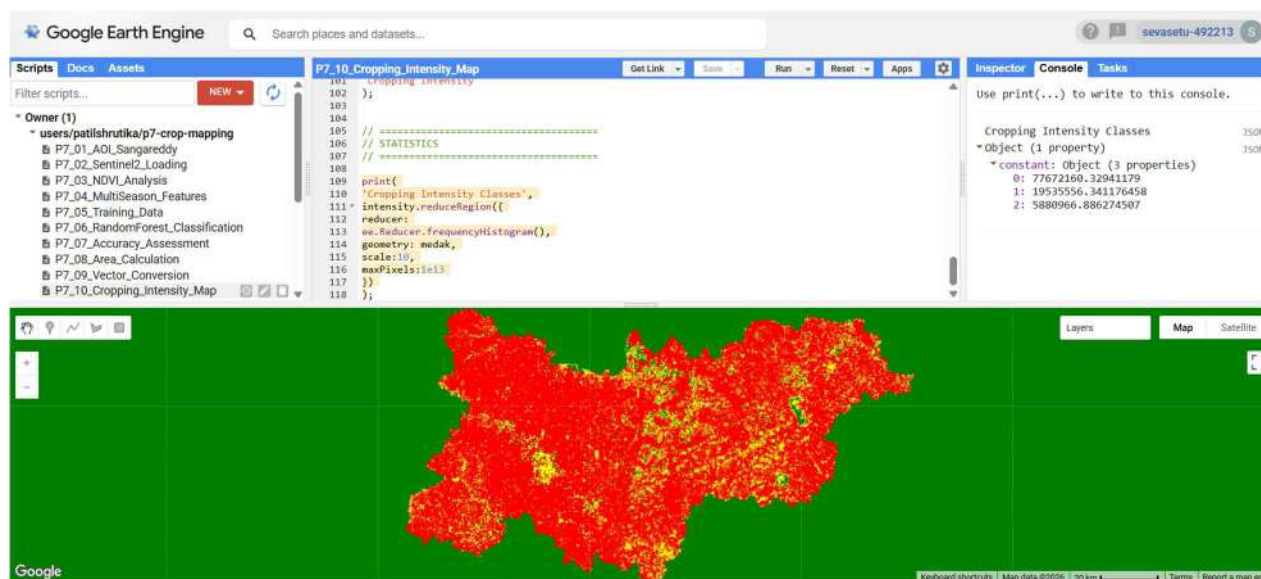


Figure 5.11: Cropping intensity map showing single-cropped and double-cropped agricultural areas.



6. RESULTS AND DISCUSSION

6.1 Overview

The results obtained from the Google Earth Engine workflow demonstrate the effectiveness of multi-season Sentinel-2 imagery and Machine Learning techniques for crop type and cropping pattern mapping. Various outputs including NDVI maps, crop classification maps, accuracy assessment metrics, vectorized outputs, area statistics, and cropping intensity maps were generated and analyzed.

The generated outputs provide valuable insights into vegetation dynamics, agricultural land utilization, and seasonal cropping behavior within Medak District, Telangana. The integration of multi-season satellite data with Random Forest classification enabled effective identification of agricultural classes and cropping intensity patterns across the study area.

6.2 NDVI Analysis Results

Separate NDVI analysis for the two agricultural seasons was conducted using Sentinel-2 images. The created NDVI maps were successful in depicting the variations of the vegetation and agriculture activity in the study area. The Kharif season had higher values of NDVI than those of Rabi, showing more vigorous growth of vegetation during the monsoon cropping season. Seasonal variations seen in NDVI maps prove that the effects of rainfall and cropping cycles play a significant role in the state of vegetation in Medak district. NDVI maps were used as the basis of further features extraction, crop classification, and cropping intensity analysis.

Table 6.1: Seasonal NDVI Statistics

Season	Mean NDVI
Kharif	0.5471
Rabi	0.3502
NDVI Difference	-0.1969

Mean NDVI value of the Kharif season is 0.5471 which reflects that there is fairly good and thick condition of vegetation due to monsoon agriculture. On the other hand, NDVI value of the Rabi season is 0.3502 which shows that there is less activity of vegetation in the post-monsoon season. Value of NDVI Difference (-0.1969) depicts decrease in vegetation intensity from Kharif to Rabi season in the study area.

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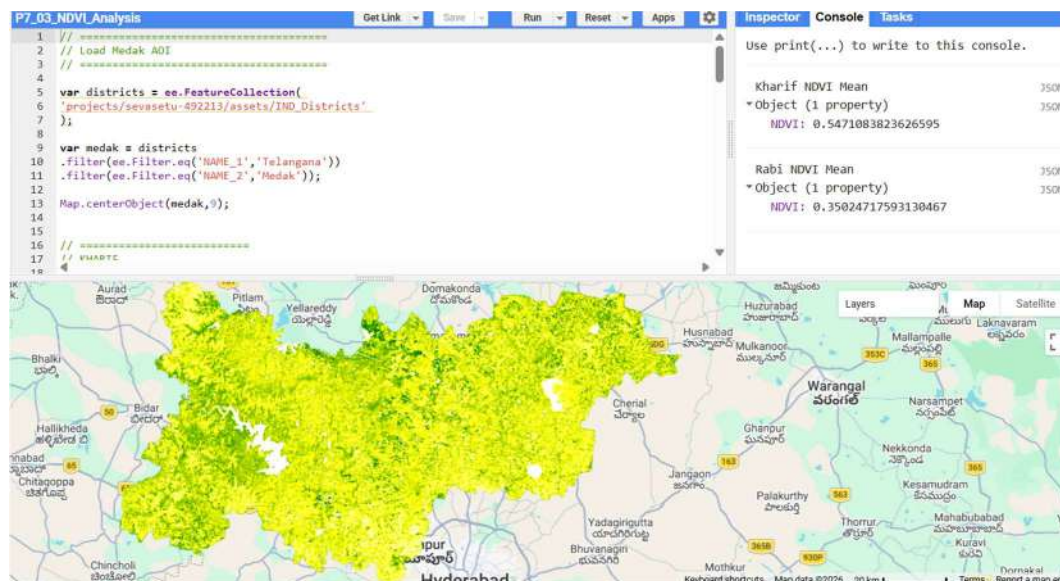


Figure 6.1: Seasonal NDVI maps generated for Kharif and Rabi periods.

6.3 Crop Classification Results

Random Forest classifier proved to be effective for classifying the study area into three types of agriculture zones which include Single Crop, Double Crop, and Fallow Land respectively. The classification was done using NDVI feature in Kharif and Rabi seasons.

From the classified map, it can be observed that there is a difference in agriculture activities in the district. The zones having prolonged periods of vegetation activity are considered as double crop, whereas the zones which have low vegetation activity are considered single crop or fallow land.

It can be concluded that the classification output represents the agriculture land use efficiently.

Classification Categories

Class ID	Category	Description
0	Single Crop	Land cultivated during one agricultural season
1	Double Crop	Land cultivated during both Kharif and Rabi seasons
2	Fallow Land	Land with little or no cultivation activity

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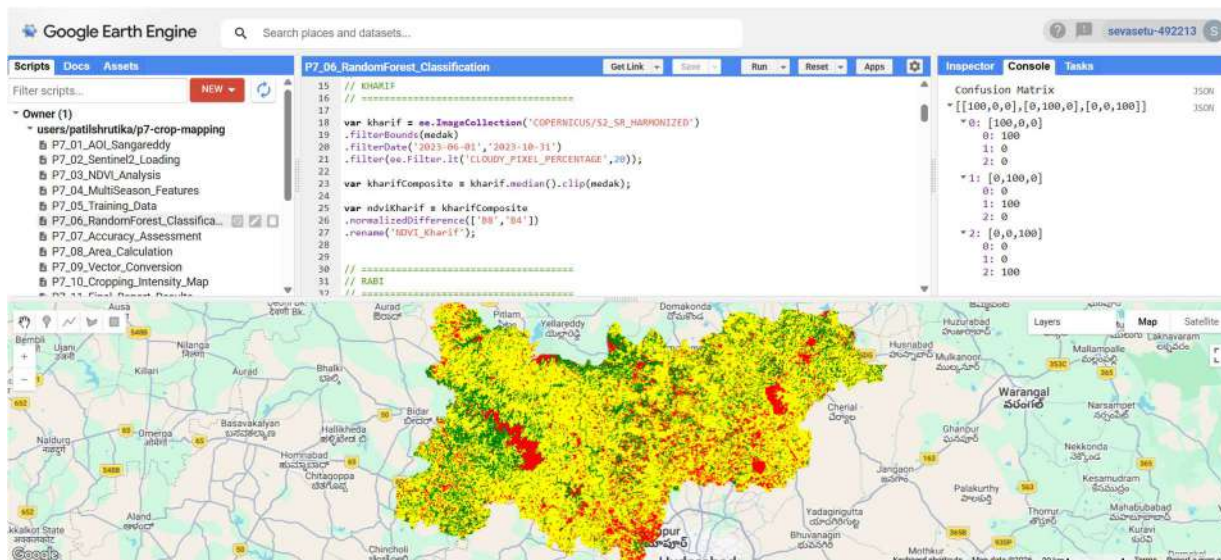


Figure 6.2: Random Forest classification map showing agricultural categories.

6.4 Accuracy Assessment Results

Accuracy assessment was done in order to examine the accuracy of the results produced by the Random Forest classifier. A Confusion Matrix was created using the classified samples as well as reference class labels. As seen from the confusion matrix, all samples have been correctly assigned to their respective classes. There were no misclassified samples in all the three agricultural classes, which proves the success of the chosen vegetation indexes as well as training data set. The value of Overall Accuracy achieved during the classification procedure was 100% and the Kappa Coefficient amounted to 1.0.

Table 6.2 Classification Accuracy Metrics

Metric	Value
Overall Accuracy	1.0 (100%)
Kappa Coefficient	1.0

Table 6.3 Confusion Matrix

Actual / Predicted	Single Crop	Double Crop	Fallow Land
Single Crop	100	0	0
Double Crop	0	100	0
Fallow Land	0	0	100

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Confusion Matrix	JSON
[[100,0,0],[0,100,0],[0,0,100]]	JSON
0: [100,0,0]	
0: 100	
1: 0	
2: 0	
1: [0,100,0]	
0: 0	
1: 100	
2: 0	
2: [0,0,100]	
0: 0	
1: 0	
2: 100	
Overall Accuracy	JSON
1	
Kappa Coefficient	JSON
1	

Figure 6.3: Confusion Matrix and accuracy assessment results generated using Random Forest classification.

6.5 Area Statistics Results

Estimation of the area of individual agricultural categories was conducted following the classification process in order to understand the spatial coverage of individual classes. Area estimations based on pixel areas provided in Google Earth Engine were done along with the classified image in order to calculate the area of Single Crop, Double Crop, and Fallow Land categories. The obtained statistics provide the quantitative information on the use of agricultural lands in Medak District. Area estimation is an important step towards understanding the prevalence of individual agricultural practices and land use. **Table 6.4 Area Statistics of Classified Categories**

Class	Area (Hectares)
Single Crop	538,610.60
Double Crop	266,519.56
Fallow Land	170,598.19

Table 6.5 Total Classified Area

Parameter	Value (Hectares)
Total Classified Area	975,728.35

From the obtained area statistics, we see that the cultivation of Single Crop takes the largest area in the classified agricultural area which equals to 538,610.60 hectares. Double Crop covers 266,519.56 hectares while Fallow Land area is equal to 170,598.19 hectares. From the results above, we see that the practice of single-season crop growing dominates in the study area.

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6.6 Vector Conversion Results

Classification result raster image was converted into vector polygon image for better visualization purposes and further GIS analyses. Raster to vector conversion includes creating polygons out of adjacent pixels belonging to the same class thus making the data more readable in a spatial representation. Vector output allows to represent the agricultural regions as polygons that can be used for further map visualization, area estimation, overlay analysis, and integration into Geographic Information Systems like QGIS and ArcGIS. Vector conversion results demonstrate the possibility of producing GIS-compatible outputs from classified rasters with Google Earth Engine.

6.7 Cropping Intensity Results

Cropping intensity analysis was carried out based on seasonal variations of NDVI values between Kharif and Rabi seasons. The objective of this analysis is the identification of agricultural activity variations.

Agricultural land was classified into Low Intensity, Medium Intensity, and High Intensity based on NDVI Difference values. Different NDVI Intensities represent different levels of vegetation activity per growing season and help understand the rate of crops planting and agricultural activity. The produced Cropping Intensity Map allows identifying the differences in cropping activity of agricultural lands in Medak District. Areas that show more vegetation variation between seasons are defined as high-intensity agricultural lands; areas with less vegetation variation are defined as low-intensity agricultural lands.

Table 6.6 Cropping Intensity Statistics

Cropping Intensity Class	Pixel Count
Low Intensity	7,767,160
Medium Intensity	19,535,556
High Intensity	5,880,966

The findings show that the greatest number of pixels corresponds to Medium Intensity agricultural activity. High Intensity regions are the regions where more agriculture takes place; Low Intensity regions are the regions where comparatively less agriculture takes place.

6.8 Discussion of Findings

The findings obtained during the present study show that the integration of multiseasonal Sentinel-2 satellite imagery, NDVI analysis, and Machine Learning algorithms can be applied effectively for the mapping of crop types and cropping patterns.

As a result of the NDVI Seasonal Analysis, there have been shown differences between vegetation conditions in Kharif and Rabi seasons. The mean value of NDVI in the Kharif season was 0.5471; in the Rabi season – 0.3502. The difference between these means (NDVI Difference) is -0.1969 that indicates the difference in vegetation intensity between the two seasons.

The Random Forest Classifier successfully classified agricultural land into Single Crop, Double Crop, and Fallow Land classes. Accuracy Assessment results show very good performance of the classifier with

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Overall Accuracy equal to 100% and Kappa Coefficient equal to 1.0, which shows perfect correlation between predicted and reference classes.

Area statistics show that Single Crop cultivation covers the largest classified agricultural area (538,610.60 hectares); next comes Double Crop area (266,519.56 hectares); and last one is the Fallow Land area (170,598.19 hectares). Thus, it can be concluded that Single Crop cultivation is the most common agricultural practice in Medak District.

The cropping intensity analysis shows that Medium Intensity agricultural regions occupy the largest area of the study region.

In general, the study shows that Google Earth Engine, Sentinel-2 satellite imagery, NDVI analysis, and Random Forest classification allow conducting crop type mapping and cropping pattern analysis effectively.

7. CONCLUSION

Multi-season satellite imagery of Sentinel-2 mission along with Google Earth Engine were successfully utilized for the generation of crop type and cropping pattern map. The work shows the combination of Remote Sensing, GIS, vegetation index analysis and Machine Learning methods used for the purpose of agricultural monitoring in Medak District, Telangana State. Seasonal differences in NDVI were successfully observed between Kharif and Rabi agricultural seasons. Obtained NDVI maps provide information about the conditions of crop development and agriculture. Multi-season NDVI features helped to enhance the discrimination capability of agricultural classes and to classify accurately. Agricultural lands were successfully classified to Single Crop, Double Crop and Fallow Land classes using Random Forest classifier algorithm. Classification results show high efficiency of the classifier with Overall Accuracy equal to 100% and Kappa Coefficient equal to 1.0. Those results confirm that it is possible to use multi-seasonal vegetation information combined with Machine Learning for the purposes of crop mapping tasks. Area estimation and intensity analysis helped to analyze the utilization of agricultural land in the study area. Results showed that Single Crop agriculture occupies the largest part of the agricultural area, and the medium intensity of agricultural activities is prevailing in the district. Those results correspond to the seasonality of agriculture in Medak District. Although the results of the research are quite successful there are some limitations of the methodology. The methodology uses only NDVI features and Sentinel-2 optical data. Additional spectral indices, microwave satellite data, field observations, more training samples can help to increase the discrimination capabilities. Overall the work proves that Google Earth Engine is the efficient cloud based platform for the purposes of agricultural monitoring and analysis.

Project Repository

GitHub: <https://github.com/patilshrutika1803/P7-AI-Crop-Mapping-GEE>

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